Statistical Analysis of Financial Time Series

Stylized Facts and Empirical Study

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1 Introduction

The statistical analysis of financial time series has become a central topic in both economics and statistical physics. In recent decades, the interdisciplinary field of econophysics has provided important insights into the behavior of financial markets, showing that asset returns exhibit robust statistical regularities that cannot be explained by classical models such as the geometric Brownian motion. These empirical regularities are commonly referred to as *stylized facts*.

The present work aims to reproduce and analyze these stylized facts using real financial data, following the framework introduced in the paper *"Statistics of Financial Time Series: Price, Returns, Volumes, Volatility"* by R. Cont (2001). The analysis is restricted to the sections of the article up to page 9, focusing on the statistical properties of asset returns.

Daily data from the S&P 500 index and the stock of UnitedHealth Group (UNH) are considered. Log-returns are computed and a set of empirical analyses is performed, including the study of autocorrelation functions, probability distributions, and scaling behavior across different time horizons.

The results obtained allow for the verification of the main stylized facts and highlight the limitations of Gaussian-based models in accurately describing financial fluctuations.

2 Theoretical background

2.1 Stylized Facts of Financial Time Series

Financial time series exhibit a number of empirical regularities that are consistently observed across different markets, assets, and time periods. These properties, known as *stylized facts*, represent fundamental characteristics of financial data that any realistic model should be able to reproduce.

One of the most important stylized facts is the **absence of autocorrelation in returns**. Empirical studies show that linear autocorrelations of asset returns are typically close to zero for all time lags. This implies that price changes are essentially unpredictable in a linear sense, which is consistent with the efficient market hypothesis.

In contrast, financial time series display a strong **volatility clustering** effect. Although returns themselves are uncorrelated, measures of volatility such as absolute returns or squared returns exhibit significant positive autocorrelation. This indicates that periods of high volatility

tend to cluster together, as do periods of low volatility, revealing the presence of temporal dependence in the magnitude of price fluctuations.

Another key feature is the presence of **heavy-tailed distributions**. The empirical distribution of returns shows a much higher probability of extreme events compared to a Gaussian distribution. This implies that large price fluctuations occur more frequently than predicted by classical models based on normality assumptions.

Finally, financial returns exhibit **aggregational normality**. As the time scale over which returns are measured increases, their distribution tends to approach a Gaussian shape. This behavior is often interpreted as a consequence of the central limit theorem, although the convergence is typically slow due to the presence of temporal dependencies such as volatility clustering.

These stylized facts provide a benchmark for evaluating financial models and highlight the limitations of simple stochastic processes in capturing the complexity of financial markets.

2.2 Returns and Log-Returns

The analysis of financial time series is typically based on asset returns rather than raw prices. Returns provide a normalized measure of price variation and allow for meaningful statistical comparisons across different assets and time periods.

Given a price time series P_t, the simple return over a time horizon τ is defined as:

$$R_\tau(t) = \frac{P_t - P_{t-\tau}}{P_{t-\tau}} \tag{1}$$

In most financial applications, it is more convenient to work with logarithmic returns (log-returns), defined as:

$$r_\tau(t) = \ln\left(\frac{P_t}{P_{t-\tau}}\right) \tag{2}$$

The case $\tau = 1$ corresponds to the standard definition of daily returns.

Log-returns present several advantages over simple returns. First, they are time-additive, meaning that returns over longer periods can be obtained by summing returns over shorter intervals:

$$r_\tau(t) = \sum_{i=1}^{\tau} r_1(t - i + 1) \tag{3}$$

This property is particularly useful when analyzing different time scales, as in the study of aggregational normality.

Moreover, for small price variations, log-returns are approximately equal to simple returns:

$$r_\tau(t) \approx R_\tau(t) \tag{4}$$

In this work, log-returns are used throughout the analysis due to their mathematical convenience and widespread use in the literature.

2.3 Statistical Tools

In order to analyze the statistical properties of financial time series, several standard statistical tools are employed throughout this work.

A central quantity is the *autocorrelation function* (ACF), which measures the linear dependence between values of a time series separated by a given lag ℓ. For a return series r_t, it is defined as:

$$\rho(\ell) = \frac{\mathbb{E}[(r_t - \mu)(r_{t+\ell} - \mu)]}{\sigma^2}, \tag{5}$$

where μ and σ^2 denote the mean and variance of the series. The ACF is used to detect the presence of linear correlations in returns.

In addition to returns, the autocorrelation of absolute returns $|r_t|$ is also considered. This quantity is commonly used as a proxy for volatility, since large absolute returns correspond to periods of high fluctuations. The persistence of autocorrelation in $|r_t|$ provides evidence of volatility clustering.

To analyze the tail behavior of return distributions, the *complementary cumulative distribution function* (CCDF) is used. For a random variable r, it is defined as:

$$P(|r| > x), \tag{6}$$

which measures the probability of observing extreme events larger than a given threshold x. This representation is particularly useful for identifying heavy-tailed behavior and approximate power-law decay.

These tools allow for a quantitative characterization of the main stylized facts observed in financial time series.

3 Data and Methodology

3.1 Data Description

The empirical analysis in this work is based on historical financial data from two assets: the S&P 500 index and the stock of UnitedHealth Group (UNH). This selection allows for a comparison between the behavior of a broad market index and that of an individual large-cap company.

The dataset covers the period from January 1, 2000 to December 31, 2025, with daily frequency. This time span provides a sufficiently long time series to capture the statistical properties of financial returns, while the daily resolution allows for the study of medium-term dynamics.

The data have been obtained using the `yfinance` Python library, which provides access to historical market data from Yahoo Finance. Daily closing prices are used as the basis for the analysis. The price series are adjusted using the Consumer Price Index (CPI), so that the analysis is performed in real terms.

It should be noted that, unlike the original study, which also considers high-frequency data, this analysis is restricted to daily observations. As a consequence, some short-term effects may appear less pronounced.

3.2 Return Computation

The computation of returns is performed using the inflation-adjusted price series described in Section 3.1. Real prices P_t^{real} are obtained by correcting nominal prices P_t with the Consumer Price Index (CPI) according to:

$$P_t^{\text{real}} = \frac{P_t}{\text{CPI}_t} \tag{7}$$

This transformation allows the analysis to be carried out in real terms, removing the effect of inflation.

Following the general definition introduced in Section 2.2, log-returns $r_\tau(t)$ are computed for different time horizons. In this work, three time scales are considered:

- $\tau = 1$: daily returns,
- $\tau = 5$: approximately weekly returns,
- $\tau = 20$: approximately monthly returns.

These time scales are used to analyze how the statistical properties of returns evolve as the observation horizon increases, particularly in the study of aggregational normality.

For $\tau = 5$ and $\tau = 20$, returns are constructed using non-overlapping intervals in order to avoid artificial correlations between consecutive observations.

Finally, for the analysis of return distributions, each return series is normalized according to:

$$r_\tau^{\text{norm}}(t) = \frac{r_\tau(t) - \mu_\tau}{\sigma_\tau} \tag{8}$$

where μ_τ and σ_τ denote the sample mean and standard deviation of the corresponding return series.

4 Empirical results

4.1 Fat Tails

One of the most prominent stylized facts of financial time series is the presence of heavy tails in the distribution of returns. This implies that extreme events occur more frequently than predicted by a Gaussian model.

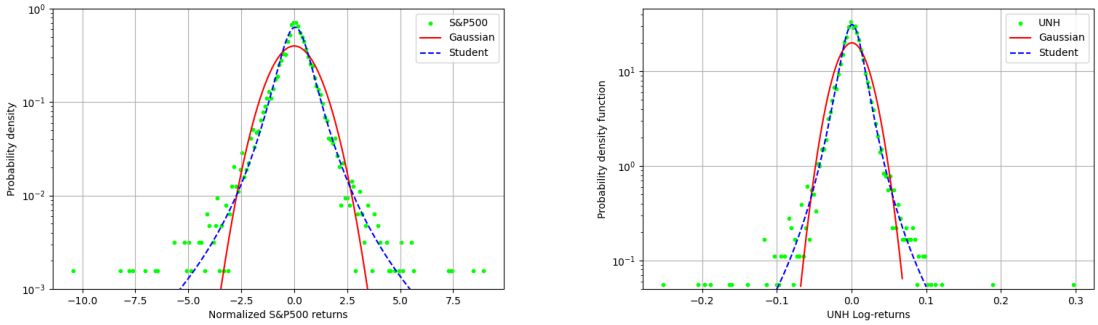

Figure 1: Probability density functions of returns. Left: normalized S&P500 returns. Right: UNH log-returns. The Gaussian distribution (red) fails to capture the tails, while the Student's t-distribution (blue) provides a better fit.

Figure 1 shows the empirical probability density functions for both assets. In both cases, the Gaussian distribution underestimates the probability of large fluctuations, while the Student's t-distribution better captures the heavy-tailed nature of the data.

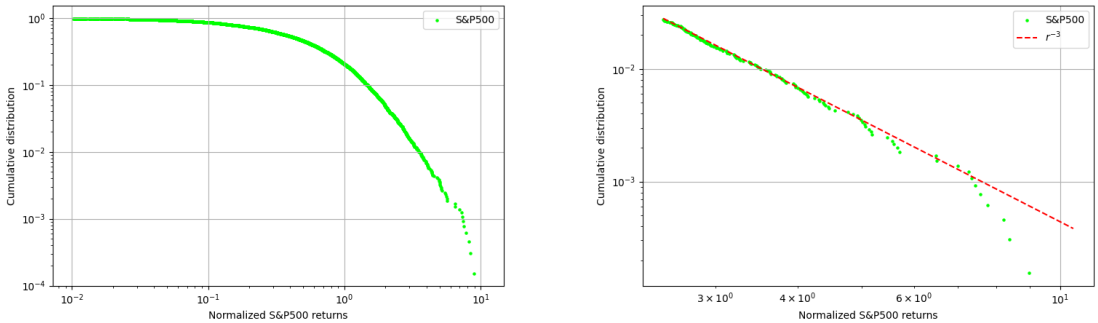

Figure 2: Complementary cumulative distribution function (CCDF) of absolute normalized S&P500 returns. Left: full range. Right: zoom on the tail region showing the approximate linear behavior in log-log scale.

To further analyze the tail behavior, the complementary cumulative distribution function (CCDF) of the absolute normalized returns is considered. As shown in Figure 2, the slow decay of the distribution indicates a higher probability of extreme events compared to a Gaussian model.

In the zoomed representation (Figure 2, right), the tail exhibits an approximately linear behavior, suggesting a power-law decay of the form:

$$P(|r| > x) \sim x^{-\alpha} \tag{9}$$

The slope is consistent with an exponent $\alpha \approx 3$, in agreement with previous empirical studies.

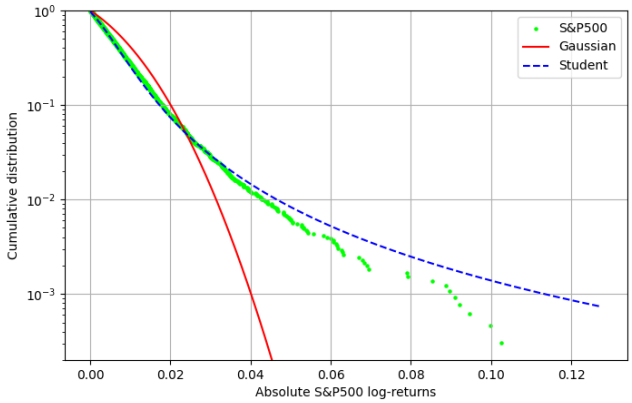

Figure 3: Comparison between the empirical CCDF and Gaussian and Student's t-distributions for the S&P500.

Finally, Figure 3 compares the empirical distribution with Gaussian and Student's t-distributions. The Gaussian model clearly underestimates tail probabilities, while the Student's distribution provides a significantly better fit, although deviations remain for extreme values.

Overall, these results confirm the presence of fat tails in financial returns and highlight the limitations of Gaussian-based models.

4.2 Autocorrelation and Volatility Clustering

Another fundamental stylized fact of financial time series is the absence of linear autocorrelation in returns, combined with the presence of significant autocorrelation in measures of volatility.

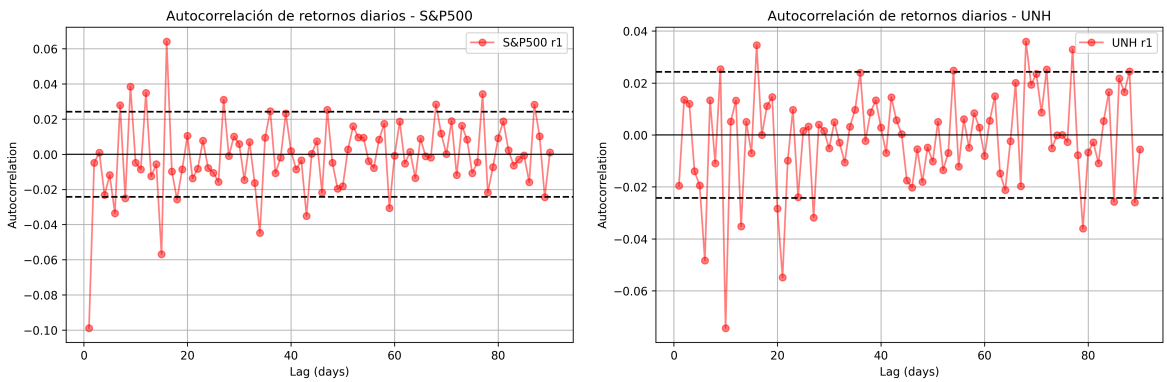

Figure 4: Autocorrelation function of daily returns. Left: S&P500. Right: UNH. Dashed lines represent confidence intervals.

Figure 4 shows the autocorrelation function (ACF) of daily returns for both assets. In both cases, the autocorrelation values fluctuate around zero and mostly remain within the confidence bands. This indicates that returns exhibit little to no linear dependence, behaving approximately as a white noise process.

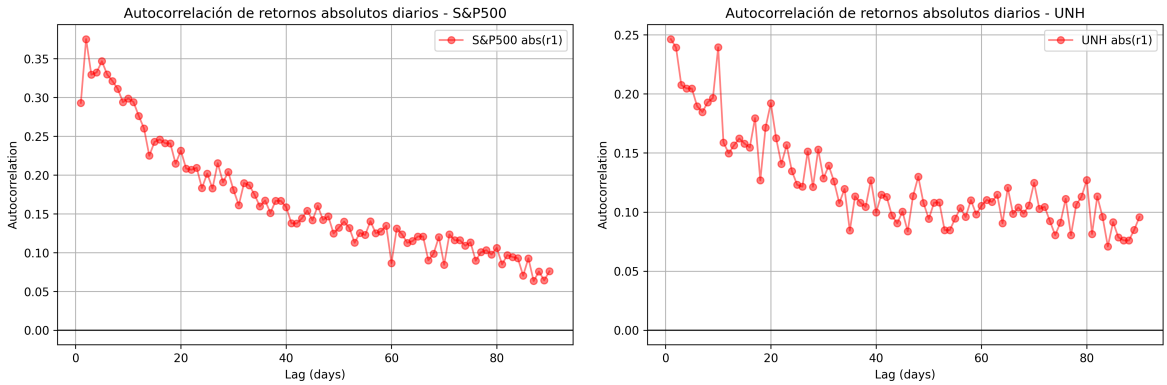

Figure 5: Autocorrelation function of absolute returns. Left: S&P500. Right: UNH.

In contrast, Figure 5 shows the autocorrelation of absolute returns, which act as a proxy for volatility. In this case, a strong positive autocorrelation is observed, decaying slowly as the lag increases. Moreover, the autocorrelation remains clearly positive for a large number of lags.

This behavior indicates the presence of volatility clustering: large price changes tend to be followed by large changes, and small changes tend to be followed by small ones. This effect introduces long-range dependence in volatility, even though returns themselves are approximately uncorrelated.

Overall, these results confirm that financial returns are consistent with the efficient market hypothesis at the level of linear correlations, while still exhibiting complex temporal structure in their volatility.

4.3 Aggregational Normality

Another well-known stylized fact of financial time series is the tendency of return distributions to approach normality as the time scale increases. This phenomenon is commonly referred to as *aggregational normality*.

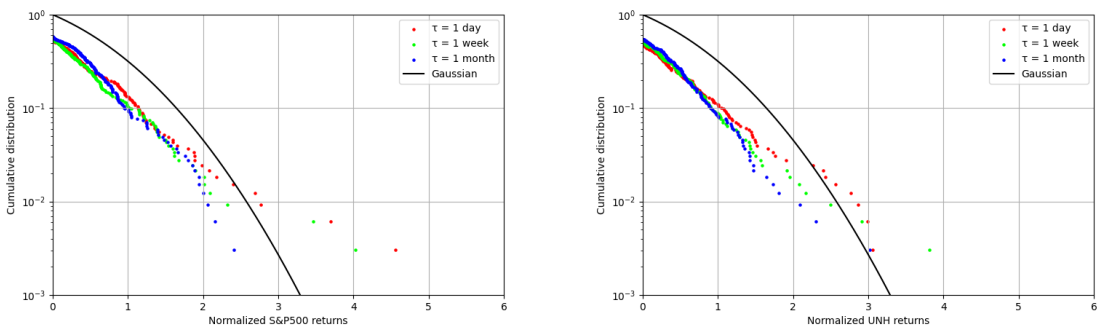

Figure 6: Complementary cumulative distribution functions (CCDF) of normalized returns for different time scales. Left: S&P500. Right: UNH.

Figure 6 shows the complementary cumulative distribution functions (CCDF) of normalized returns for different time horizons: $\tau = 1$ day, $\tau = 5$ days, and $\tau = 20$ days. For comparison, the Gaussian distribution is also plotted.

As the time scale increases, the empirical distributions become progressively closer to the Gaussian curve. In particular, the tails of the distribution become less pronounced, indicating a reduction in extreme fluctuations at larger aggregation levels.

However, even at $\tau = 20$, noticeable deviations from the Gaussian distribution remain, especially in the tail region. This suggests that the convergence towards normality is slow and incomplete at the daily time scale considered in this study.

This behavior is consistent with the presence of volatility clustering. Since volatility exhibits long-range dependence, the central limit theorem does not apply straightforwardly, and the aggregation of returns does not immediately lead to a Gaussian distribution.

Overall, these results confirm that financial returns exhibit a gradual, but slow, convergence towards normality as the time scale increases, in agreement with the empirical findings reported in the literature.

5 Discussion and Conclusions

The empirical analysis carried out in this work reproduces the main stylized facts of financial time series described in the reference paper. Using daily data from the S&P500 index and UnitedHealth Group (UNH), several robust statistical properties of financial returns have been identified.

First, the empirical distributions of returns show clear deviations from the Gaussian model. In particular, the probability of large fluctuations is significantly higher than predicted by a normal distribution. This behavior is observed both in the probability density functions and in the complementary cumulative distributions. The Student's t-distribution provides a better description of the empirical data, although deviations remain for the most extreme observations. The log-log representation of the S&P500 tail also suggests an approximate power-law behavior, with an exponent close to $\alpha \approx 3$, in qualitative agreement with the empirical evidence reported in the literature.

Second, the autocorrelation analysis confirms one of the central stylized facts of financial markets: returns themselves exhibit little linear dependence. For both assets, the autocorrelation of daily returns fluctuates around zero and remains mostly within the confidence bands. This indicates that the direction of price changes is difficult to predict using linear correlations alone, which is consistent with the idea of informational efficiency in financial markets.

However, the behavior changes when absolute returns are considered. The autocorrelation of $|r_t|$, used as a proxy for volatility, remains positive and decays slowly with the lag. This reveals the presence of volatility clustering: periods of large fluctuations tend to be followed by further periods of large fluctuations, while calmer periods also tend to persist. Therefore, although returns are approximately uncorrelated, their magnitude contains temporal structure. This distinction is essential, since it shows that financial time series are not simply independent random processes.

The analysis of different time scales also provides evidence of aggregational normality. When returns are computed over longer horizons, from daily to approximately weekly and monthly scales, the empirical distributions tend to move closer to the Gaussian curve. Nevertheless, the convergence is slow and incomplete, especially in the tails. This result is consistent with the

presence of volatility clustering and heavy-tailed behavior, which prevent a direct and rapid convergence to normality.

Some limitations of the analysis should also be noted. The original paper includes high-frequency data, whereas this work is restricted to daily observations. As a consequence, some short-term effects may appear less pronounced or cannot be reproduced exactly. In addition, the comparison between the S&P500 and UNH shows that individual stocks can display more irregular behavior than broad market indices, particularly in the tails and at longer aggregation horizons.

Overall, the results confirm that the main stylized facts of financial time series are present in the analyzed data: heavy-tailed return distributions, absence of linear autocorrelation in returns, volatility clustering, and slow convergence towards Gaussian behavior at larger time scales. These findings highlight the limitations of simple Gaussian-based models and motivate the use of more sophisticated approaches capable of capturing nonlinear dependence, time-varying volatility, and extreme events in financial markets.